

FUNDAMENTAL CAUSES OF GROWTH

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CAUSES OF GROWTH

Research on income differences can arguably be classified into one or more arrows in the following chain of causality:

Geography, Climate, Luck $\Rightarrow$ Human Capital, Physical Capital, TFP $\Rightarrow$ Income

$\Downarrow$

Institutions, Culture $\Rightarrow$ Human Capital, Physical Capital, TFP $\Rightarrow$ Income

$\Downarrow$

Policies, Rule of Law, Corruption $\Rightarrow$ Human Capital, Physical Capital, TFP $\Rightarrow$ Income

- Development accounting is about right-most arrows
- In this lecture, we discuss research on the arrows to the left

Source: Hsieh and Klenow (2010)

CROSS-COUNTRY GROWTH REGRESSIONS

- 1990s saw a flourishing of research on correlates of growth
- Prominent studies by Barro (1991), Sala-i-Martin (1997)
- Some people interpreted these correlations as causal
- This gave “cross-country growth regressions” a bad name

Regressions for per capita growth rate

Independent variable	(1)	(2)
Log(GDP)	-.0254 (.0031)	-.0225 (.0032)
Male secondary and higher schooling	.0118 (.0025)	.0098 (.0025)
Log(life expectancy)	.0423 (.0137)	.0418 (.0139)
Log(GDP) * male schooling	-.0062 (.0017)	-.0052 (.0017)
Log(fertility rate)	-.0161 (.0053)	-.0135 (.0053)
Government consumption ratio	-.136 (.026)	-.115 (.027)
Rule of law index	.0293 (.0054)	.0262 (.0055)
Terms of trade change	.137 (.030)	.127 (.030)
Democracy index	.090 ^a (.027)	.094 (.027)
Democracy index squared	-.088 (.024)	-.091 (.024)
Inflation rate	-.043 (.008)	-.039 (.008)
Sub-Saharan Africa dummy		-.0042 ^b (.0043)
Latin America dummy		-.0054 (.0032)
East Asia dummy		.0050 (.0041)
R ²	.58, .52, .42	.60, .52, .47
Number of observations	80, 87, 84	80, 87, 84

Source: Barro (1998)

CROSS-COUNTRY GROWTH REGRESSIONS

- Cross-country growth regressions hard to interpret
- Reverse causation: Growth may cause other things like democracy / rule of law / government spending
 - This is called “modernization theory”
- Omitted variables bias: Both growth and variables of interest may be caused by a third factor
- Theory suggests that many factors have level effects as opposed to growth effects (but transition dynamics are slow)

FUNDAMENTAL CAUSES OF GROWTH

- Many candidate causes:
 - Geography, trade, institutions
 - Luck (e.g., leaders), culture, religion, etc.
- To provide convincing evidence on fundamental causes of growth, we need exogenous variation in candidate causes
- Subsequent literature includes many creative approaches to this challenging task
- Here we will sample a bit from this (large) literature

INSTITUTIONS AS A FUNDAMENTAL CAUSE OF GROWTH

- Long-standing literature arguing that good institutions cause high levels of income
 - Secure property rights, rule of law, constraints on the executive, etc.
- Important work along these lines: North and Thomas (1973), Jones (1981), North and Weingast (1989), North (1990), Engerman and Sokoloff (1997)
- Literature mostly qualitative before late 1990s
- Where can we find exogenous variation in institutions?
 - A few examples: North vs. South Korea, East vs. West Germany
 - But something more systematic?

COLONIAL ORIGINS

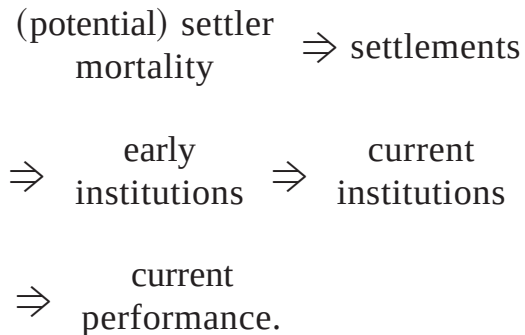
- Acemoglu, Johnson, Robinson (2001) proposed a theory of institutional differences among countries colonized by Europeans
- Conditions in the colonies led colonizers to create different institutions in different colonies
- Where colonizers settled, they set up “good” institutions (secure property rights, rule of law, constraints on the executive)
- Where they didn’t settle, they set up “extractive” institutions
- These institutions have persisted and have affected growth in these countries

INCREDIBLY CONTROVERSIAL PAPER

- Some: Colonial Origins paper shouldn't get A in an undergrad class
- Others: Nobel Prize!
- Case against:
 - Ideas not original / Data low quality / Identifying restriction implausible / Questionable manipulation of data / Standard errors are wrong
- Case for:
 - Formal analysis using data and statistical methods constitute huge improvement in discipline imposed on analysis
 - Traditional historical analysis vague, undisciplined (pick and choose anecdotes to fit the story) and hard to replicate
 - Fact that it is easy to critique is a consequence of clarity of methods
 - Pioneered causal historical analysis, which has been very influential

SETTLER MORTALITY AS AN INSTRUMENT

- To test their theory, AJR propose using settler mortality as an instrument for current institutions
- Basic idea:
 - Colonizers didn't settle where settler mortality was high
 - Settler mortality was determined by disease environment (mostly malaria and yellow fever)
 - Places with malaria and yellow fever got worse institutions



Source: Acemoglu, Johnson, Robinson (2001)

- Second stage regression:

$$\log y_i = \mu + \alpha R_i + \mathbf{X}_i' \gamma + \epsilon_i$$

- First state regression:

$$R_i = \zeta + \beta \log M_i + \mathbf{X}_i' \delta + v_i$$

- $\log y_i$ is log GDP per capita in 1995 PPP adjusted
- R_i is protection against expropriation from Political Risk Services
(Is this institutions or policy? Glaeser, et al. (2004))
- $\log M_i$ is log settler mortality

IS SETTLER MORTALITY A VALID INSTRUMENT?

- For settler mortality to be a valid instrument, we must have:
 - Relevance: It must strongly predict current institutions
 - Exclusion: It must not affect current output through any other channel than current institutions (conditional on controls)
- What might be an important threat to the exclusion restriction?

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- For settler mortality to be a valid instrument, we must have:
 - Relevance: It must strongly predict current institutions
 - Exclusion: It must not affect current output through any other channel than current institutions (conditional on controls)
- What might be an important threat to the exclusion restriction?
- Settler mortality likely correlates with current disease environment which may have a direct effect on GDP per capita today
(Bloom and Sachs 98, Gallup and Sachs 98, Sachs and Malaney 02, Alsan 15, Dupras 26)

IS SETTLER MORTALITY A VALID INSTRUMENT?

- AJR argue that native population had developed immunity to malaria and yellow fever
- Settlers died but natives didn't (after early childhood)
- Yellow fever has been largely eradicated since
- AJR argue: “these diseases are therefore unlikely to be the reason why many countries in Africa and Asia are very poor today”
- Counterargument: Even if mortality is not high, morbidity is substantial (also TseTse fly)

IS SETTLER MORTALITY A VALID INSTRUMENT?

- Another threat: Settlers brought themselves and their human capital
- Hard to distinguish human capital and institutions stories
- Glaeser et al. (2004) argue for human capital story

SETTLER MORTALITY VARIABLE

- Largely from work of Philip Curtin (Curtin 89, 98)
- Not actually mortality of settlers
- Mostly mortality of soldiers
- For Latin American mortality of bishops from Gutierrez (1986)
rescaled for comparability with soldiers
- More on this in a few slides

- “Reduced form” regression:

$$\log y_i = \mu + \lambda \log M_i + \mathbf{X}_i' \gamma + \epsilon_i$$

- First state regression:

$$R_i = \zeta + \beta \log M_i + \mathbf{X}_i' \delta + v_i$$

- IV estimate is the ratio of λ and β

FIRST STAGE

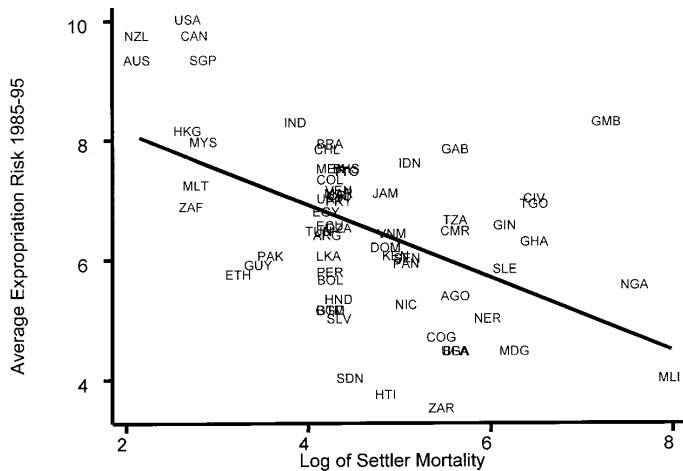


FIGURE 3. FIRST-STAGE RELATIONSHIP BETWEEN SETTLER MORTALITY AND EXPROPRIATION RISK

Source: Acemoglu, Johnson, Robinson (2001)

REDUCED FORM

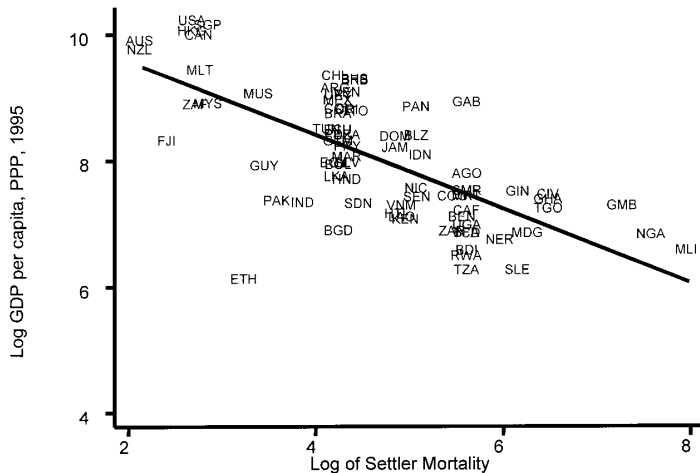


FIGURE 1. REDUCED-FORM RELATIONSHIP BETWEEN INCOME AND SETTLER MORTALITY

Source: Acemoglu, Johnson, Robinson (2001)

TABLE 4—IV REGRESSIONS OF LOG GDP PER CAPITA

	Base sample (1)	Base sample (2)	Base sample without Neo-Europes (3)	Base sample without Neo-Europes (4)	Base sample without Africa (5)	Base sample without Africa (6)	Base sample with continent dummies (7)	Base sample with continent dummies (8)	Base sample, dependent variable is log output per worker (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk 1985–1995	0.94 (0.16)	1.00 (0.22)	1.28 (0.36)	1.21 (0.35)	0.58 (0.10)	0.58 (0.12)	0.98 (0.30)	1.10 (0.46)	0.98 (0.17)
Latitude		−0.65 (1.34)		0.94 (1.46)		0.04 (0.84)		−1.20 (1.8)	
Asia dummy							−0.92 (0.40)	−1.10 (0.52)	
Africa dummy							−0.46 (0.36)	−0.44 (0.42)	
“Other” continent dummy							−0.94 (0.85)	−0.99 (1.0)	
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995									
Log European settler mortality	−0.61 (0.13)	−0.51 (0.14)	−0.39 (0.13)	−0.39 (0.14)	−1.20 (0.22)	−1.10 (0.24)	−0.43 (0.17)	−0.34 (0.18)	−0.63 (0.13)
Latitude		2.00 (1.34)		−0.11 (1.50)		0.99 (1.43)		2.00 (1.40)	
Asia dummy							0.33 (0.49)	0.47 (0.50)	
Africa dummy							−0.27 (0.41)	−0.26 (0.41)	
“Other” continent dummy							1.24 (0.84)	1.1 (0.84)	
R ²	0.27	0.30	0.13	0.13	0.47	0.47	0.30	0.33	0.28
Panel C: Ordinary Least Squares									
Average protection against expropriation risk 1985–1995	0.52 (0.06)	0.47 (0.06)	0.49 (0.08)	0.47 (0.07)	0.48 (0.07)	0.47 (0.07)	0.42 (0.06)	0.40 (0.06)	0.46 (0.06)
Number of observations	64	64	60	60	37	37	64	64	61

Source: Acemoglu, Johnson, Robinson (2001)

QUANTITATIVELY LARGE

- Consider two “typical countries”: Nigeria and Chile
(typical in that they are virtually on the regression line)
- Differ by 2.24 in expropriation risk
- Fitted difference in $0.94 \times 2.24 = 2.06$ in logs or 7-fold in levels
- Actual difference: 11-fold in levels

TABLE 5—IV REGRESSIONS OF LOG GDP PER CAPITA WITH ADDITIONAL CONTROLS

	Base sample (1)	Base sample (2)	British colonies only (3)	British colonies only (4)	Base sample (5)	Base sample (6)	Base sample (7)	Base sample (8)	Base sample (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk, 1985–1995	1.10 (0.22)	1.16 (0.34)	1.07 (0.24)	1.00 (0.22)	1.10 (0.19)	1.20 (0.29)	0.92 (0.15)	1.00 (0.25)	1.10 (0.29)
Latitude		−0.75 (1.70)				−1.10 (1.56)		−0.94 (1.50)	−1.70 (1.6)
British colonial dummy	−0.78 (0.35)	−0.80 (0.39)							
French colonial dummy	−0.12 (0.35)	−0.06 (0.42)							0.02 (0.69)
French legal origin dummy					0.89 (0.32)	0.96 (0.39)			0.51 (0.69)
<i>p</i> -value for religion variables							[0.001]	[0.004]	[0.42]
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995									
Log European settler mortality	−0.53 (0.14)	−0.43 (0.16)	−0.59 (0.19)	−0.51 (0.14)	−0.54 (0.13)	−0.44 (0.14)	−0.58 (0.13)	−0.44 (0.15)	−0.48 (0.18)
Latitude		1.97 (1.40)				2.10 (1.30)		2.50 (1.50)	2.30 (1.60)
British colonial dummy	0.63 (0.37)	0.55 (0.37)							
French colonial dummy	0.05 (0.43)	−0.12 (0.44)							−0.25 (0.89)
French legal origin					−0.67 (0.33)	−0.7 (0.32)			−0.05 (0.91)
<i>R</i> ²	0.31	0.33	0.30	0.30	0.32	0.35	0.32	0.35	0.45
Panel C: Ordinary Least Squares									
Average protection against expropriation risk, 1985–1995	0.53 (0.19)	0.47 (0.07)	0.61 (0.09)	0.47 (0.06)	0.56 (0.06)	0.56 (0.06)	0.53 (0.06)	0.47 (0.06)	0.47 (0.06)
Number of observations	64	64	25	25	64	64	64	64	64

Source: Acemoglu, Johnson, Robinson (2001)

ALBOUY (2012) CRITIQUE

- Mortality rates for only 28 out of 64 countries from within country
- Lots of choices as to which neighboring country to use
- Seven countries (all over Africa) get (different) rates from campaigns that occurred in Mali

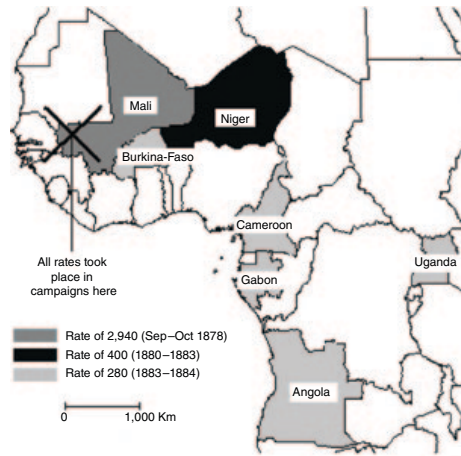


FIGURE 1. ASSIGNMENT OF MORTALITY RATES FROM MALI

ALBOUY (2012) CRITIQUE

- Bishop rates for 16 Latin American countries based on 4, 5, and 10 deaths out of at-risk populations of 24, 28.5, and 30.5 bishops in three regions
- Rates for three regions not statistically significantly different from each other or different from similar rates in Europe
- Multiplied by 4.25 to benchmark with mortality of French soldiers in Mexico in 1862-3

ALBOUY (2012) CRITIQUE

- Use campaign rates rather than barracks rates in some cases

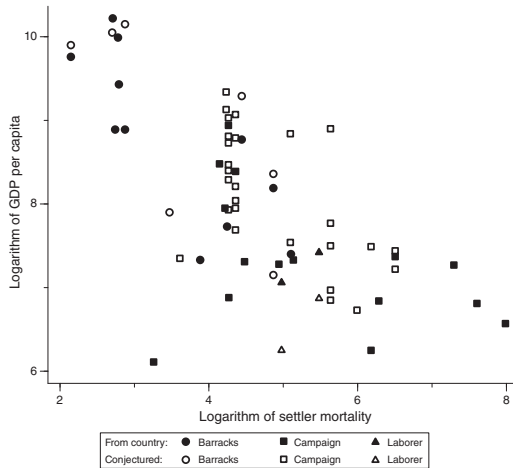
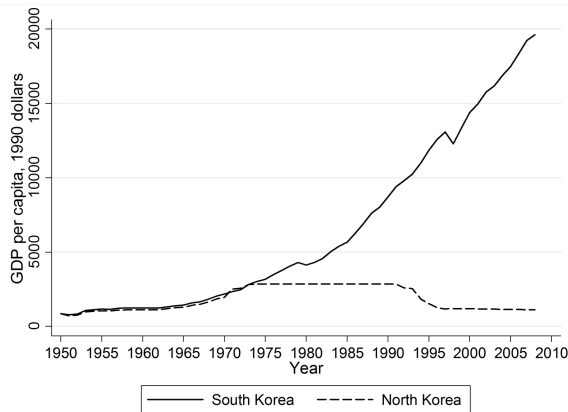


FIGURE 2B. INCOME PER CAPITA AND SETTLER MORTALITY ACCORDING TO MORTALITY RATE CHARACTERISTICS

- Results robust to lots of variation in choices made
- Can cap all mortality rates at 250
- Results robust to various ways of benchmarking bishop data
- Albouy's preferred results largely driven by Gambia
- Little difference between activities of soldiers on campaign and in barracks

INSTITUTIONS OR JUST LUCK?

- Much work on institutions and growth
- Prominent example: South Korea vs. North Korea
- How convincing is this?
- Might this be luck?



Source: Daron Acemoglu.

SOUTH KOREA VS. NORTH KOREA



Source: National Geographic

SOUTH KOREA VS. NORTH KOREA

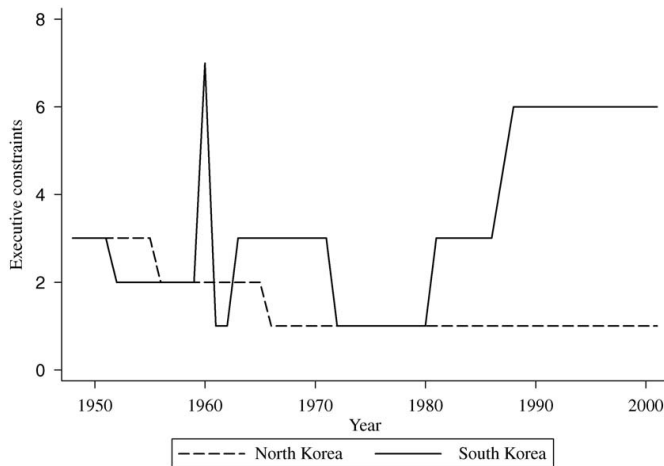


Figure 1. Executive constraints 1948–2001 North versus South Korea.

Source: Glaeser, La Porta, Lopez-de-Silanes, Shleifer (2004)

INSTITUTIONS OR LUCK?

- Gleaser et al. (2004) argue that institutions view doesn't work for Korea
 - South Korea grew rapidly under one-party dictatorship
 - Only improved institutions (constraints on executive) **after** long period of high growth
- Alternative: South Korea was lucky to have a “good” leader
 - Growth took off under leadership of Park Chung-hee (1961-1979)
 - He was not constrained by “good institutions”
- Perhaps which countries develop and which don't comes down to luck regarding leaders

DO LEADERS MATTER?

- Views differ sharply!
- Tolstoy: Historical figures mere ex post justifications for events out of any individual's influence
- Marxists: "Materialist dialectic" holds that social and economic forces trump individuals
- John Keegan: The political history of the twentieth century can be found in the biographies of six men: Lenin, Stalin, Hitler, Mao, Roosevelt and Churchill.

DO LEADERS MATTER?

- Claim: National leaders cause changes in growth
- Simple minded evidence: Look at changes in growth when leaders change

DO LEADERS MATTER?

- Claim: National leaders cause changes in growth
- Simple minded evidence: Look at changes in growth when leaders change
- But leadership transitions are non-random
 - Bad shocks may cause leaders to lose power
 - Good shocks may shield leaders

- Look at leadership transition due to leaders dying in office
- Consider cases when leaders die of natural causes or due to accidents (not assassinations)
- **Timing** of transition is random

LEADER TRANSITIONS

TABLE I
HOW LEADERS LEAVE POWER

130 Countries							
All Leaders from 1945 or National Independence Date through 2000							
Number of Observations, by Type							
Lost election 310	Term limits 178		Voluntary retirement 131	Deposed 222	Death ^a 105		Other 225
Assassination 28		Natural 65			Accidental 12		105
Heart disease 29	Cancer 12	Stroke 6	Other disease 6	Surgical complications 3	Other 9	Air crash 8	Other 4
							77 ^c

a. There are 21 further cases (not included here) where leaders are killed during a coup.

b. There are 1294 distinct terms in which leaders are in power in the data set, but only 1184 counted in this table, as we do not witness the exit of leaders who are still in power at the end of the year 2000.

c. There are 77 cases of leaders who die in office by natural causes or accidents, but only 57 who die during periods where there are available growth data before and after the leader's death.

Source: Jones and Olken (2005)

DRAMATIC CASES

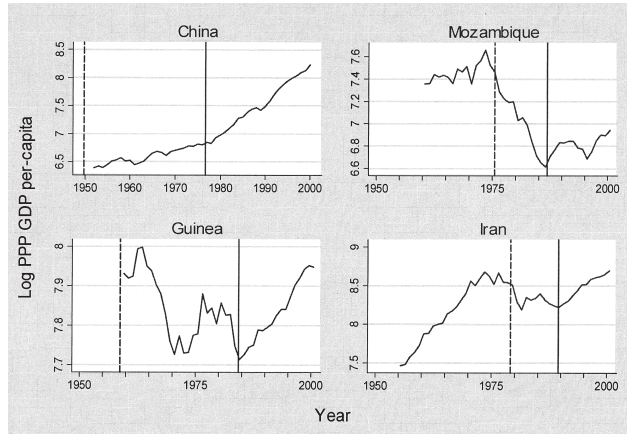


FIGURE I

Growth and Leader Deaths

Source: Jones and Olken (2005). China: Mao. Mozambique: Samora Machel (nationalized private land).

Guinea: Sekou Toure (totalitarianism/violent purges). Iran: Ayatollah Khomeini (Iran-Iraq war).

HOW TO MEASURE EFFECT?

- Timing of death is random, but timing of accession to power is not
- Can't use growth during leader's reign
- Calculate growth in T years before and after random transition

MODEL OF GROWTH

$$g_{it} = \alpha_z \text{PRE}_z + \beta_z \text{POST}_z + v_i + v_t + \epsilon_{it}$$

- z denotes particular leader death
- PRE_z : dummy for T years prior to death
- POST_z : dummy for T years after death (they exclude year of death)
- v_i, v_t : country and year fixed effects
- ϵ_{it} : Other influences on growth

- Wald test:

$$J = \frac{1}{Z} \sum_{z=1}^Z \frac{(\widehat{\text{POST}} - \text{PRE}_z)^2}{2\hat{\sigma}_{\epsilon z}^2/T}$$

- Compares change in growth pre and post leader death to usual variation in growth
- Under null of no effect, $Z \times J$ will be distributed $\chi^2(Z)$
- $\widehat{\text{POST}} - \text{PRE}_z = \alpha_z - \beta_z$ from regression on previous page

- Rank test:

$$K = \frac{\sum (y_z - 1/4)}{\sqrt{Z/48}}$$

- $y_z = |r_z - 1/2|$ where r_z is rank of $\widehat{\text{POST}} - \text{PRE}_z$ among all $\widehat{\text{POST}} - \text{PRE}_{it}$
- r_z is $U[0,1]$ under null that leaders don't matter. So, $E[y_z] = 1/4$ and $\text{var}[y_z] = 1/48$

DO LEADERS MATTER?

TABLE III
DO LEADERS MATTER?

	All leaders			Leaders with tenure ≥ 2 years		
	<i>J</i> - statistic	Wald <i>P</i> -value	Rank <i>P</i> -value	<i>J</i> - statistic	Wald <i>P</i> -value	Rank <i>P</i> -value
Treatment timings						
t	1.312	.0573*	0.017**	1.392	.0390**	0.004***
$t + 1$	1.272	.0845*	0.075*	1.361	.0537*	0.052*
$t + 2$	1.308	.0669*	0.172	1.443	.0314**	0.121
Control timings						
$t - 5$	0.841	.7953	0.446	0.918	.6269	0.357
$t - 6$	0.986	.5026	0.806	0.962	.5409	0.905
Number of leaders (t)	57	57	57	47	47	47
Number of observations (t)	5567	5567	5567	5567	5567	5567

Source: Jones and Olken (2005).

MAGNITUDE OF EFFECT

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- A few more assumptions imply that a one standard deviation increase in leader quality increases growth by 1.47 percentage points per year
- Huge effect!!

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- A few more assumptions imply that a one standard deviation increase in leader quality increases growth by 1.47 percentage points per year
- Huge effect!!
- For autocracies, effect is 2.1 percentage points per year
- For democracies, effect is zero

WHEN DO LEADERS MATTER?

TABLE V
INTERACTIONS WITH TYPE OF POLITICAL REGIME IN YEAR PRIOR TO DEATH

	<i>J</i> - statistic	Wald <i>P</i> -value	Rank <i>P</i> -value	<i>J</i> - statistic	Wald <i>P</i> -value	Rank <i>P</i> -value
	Autocrats (Polity IV)			Democrats (Polity IV)		
Treatment timings						
<i>t</i>	1.621	0.019**	0.040**	1.000	0.460	0.106
<i>t</i> + 1	1.672	0.016**	0.017**	0.932	0.552	0.712
<i>t</i> + 2	1.592	0.028**	0.051*	1.021	0.432	0.636
Control timings						
<i>t</i> - 5	0.849	0.698	0.837	0.866	0.632	0.075*
<i>t</i> - 6	1.094	0.334	0.977	0.647	0.873	0.191
Number of leaders (<i>t</i>)	29	29	29	22	22	22

Source: Jones and Olken (2005). One standard deviation increase in autocratic leader quality increases growth by 2.1 percentage points per year.

IS IT REALLY ALL JUST POLITICS?

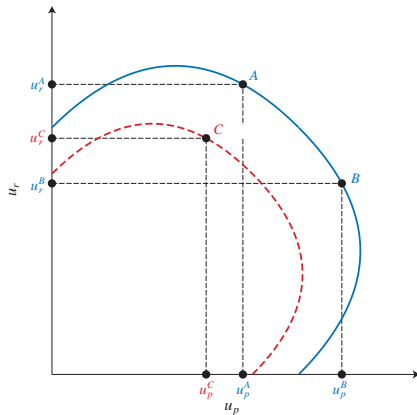
- One view: We know what policies and institutions result in prosperity
 - Rule of law and secure property rights
 - Competitive markets
 - Responsible monetary and fiscal policy
- Why don't we have these institutions and policies everywhere?
- Because reforms are blocked by powerful interest groups (the elite)

THE COASE THEOREM

- **Coase Theorem:** *Absent transaction costs, private negotiations between agents will result in resources being efficiently allocated independent of how the rights over the resources are initially allocated.*
- If a vastly more efficient/productive outcome is possible, why don't the elite and the rest of the population strike a bargain where the elite captures enough of the surplus to agree to the reform?

PROBLEM 1: ASYMMETRIC INFORMATION

- Elite and population have conflicting interests
- They also have imperfect information about each other's resolve / strength
- This can result in bargaining being delayed
- Each side is trying to force other to concede
- Alesina and Drazen (1991):
Why Are Stabilizations Delayed?



Source: Acemoglu (2025) (adapted)

PROBLEM 2: INSUFFICIENT FISCAL INSTRUMENTS

- Reforms may undermine elite privilege (e.g., reduce monopoly rents)
- Benefit population but hurt elite ($C \rightarrow B$)
- In principle, elite could be compensated ($B \rightarrow A$)
- But adequate fiscal tools may not exist
- Elite blocks reform because it would be an *economic loser*

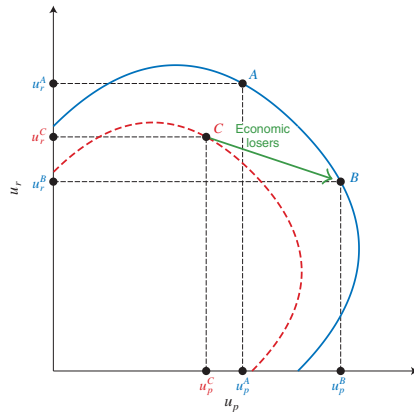


FIGURE 8. ECONOMIC LOSERS

Source: Acemoglu (2025)

PROBLEM 3: NO COMMITMENT

- If reforms allow population to prosper, it will become more politically powerful
- Elite blocks reforms because it would be a *political loser*
- Fundamental problem: population cannot commit not to wield its political power
- Acemoglu (2003): There Is No Political Coase Theorem

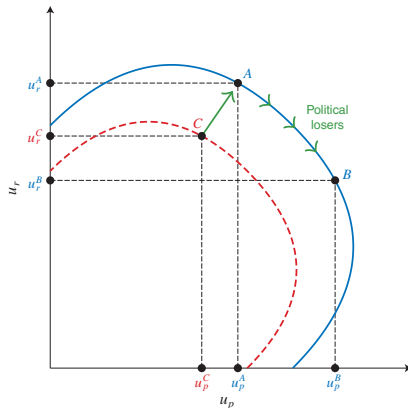
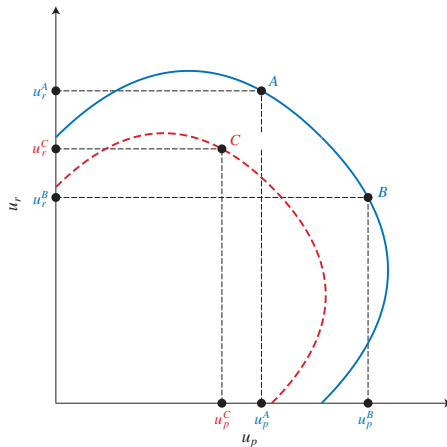


FIGURE 9. POLITICAL LOSERS

Source: Acemoglu (2025)

PROBLEM 4: NO COMMITMENT / HOLDUP

- Elite would like to promise secure property rights
- Would raise tax revenue accruing to elite
- But elite cannot commit to maintain secure property rights ex post (once population has invested)
- North and Weingast (1989): Constitutions and Commitment



Source: Acemoglu (2025) (adapted)